

# ***Concept of Prime no***

***prime no: A number that has exactly two factors  
1 and itself.***

**examples:**

**N = 11**       factors 1 and 11 only

**N = 12**        
factors 1,2,3,4,6,12.so this is not a  
prime number.

# Brute force approach:

```
9  #include <bits/stdc++.h>
10 using namespace std;
11
12 bool isPrimeNo(int N)
13 {
14     int cnt = 0;
15
16     for (int i = 1; i <= N; i++)
17     {
18         if (N % i == 0)
19         {
20             cnt++;
21         }
22     }
23     if(cnt!=2)
24     {
25         return false;
26     }
27     return true;
28 }
29
30 int main()
31 {
32     int N = 11;
33     if (isPrimeNo(N))
34     {
35         cout << "It's a prime number." << endl;
36     }
37     else
38     {
39         cout << "It's not a prime number." << endl;
40     }
41
42     return 0;
43 }
```

It's a prime number.

Time complexity: $O(N)$

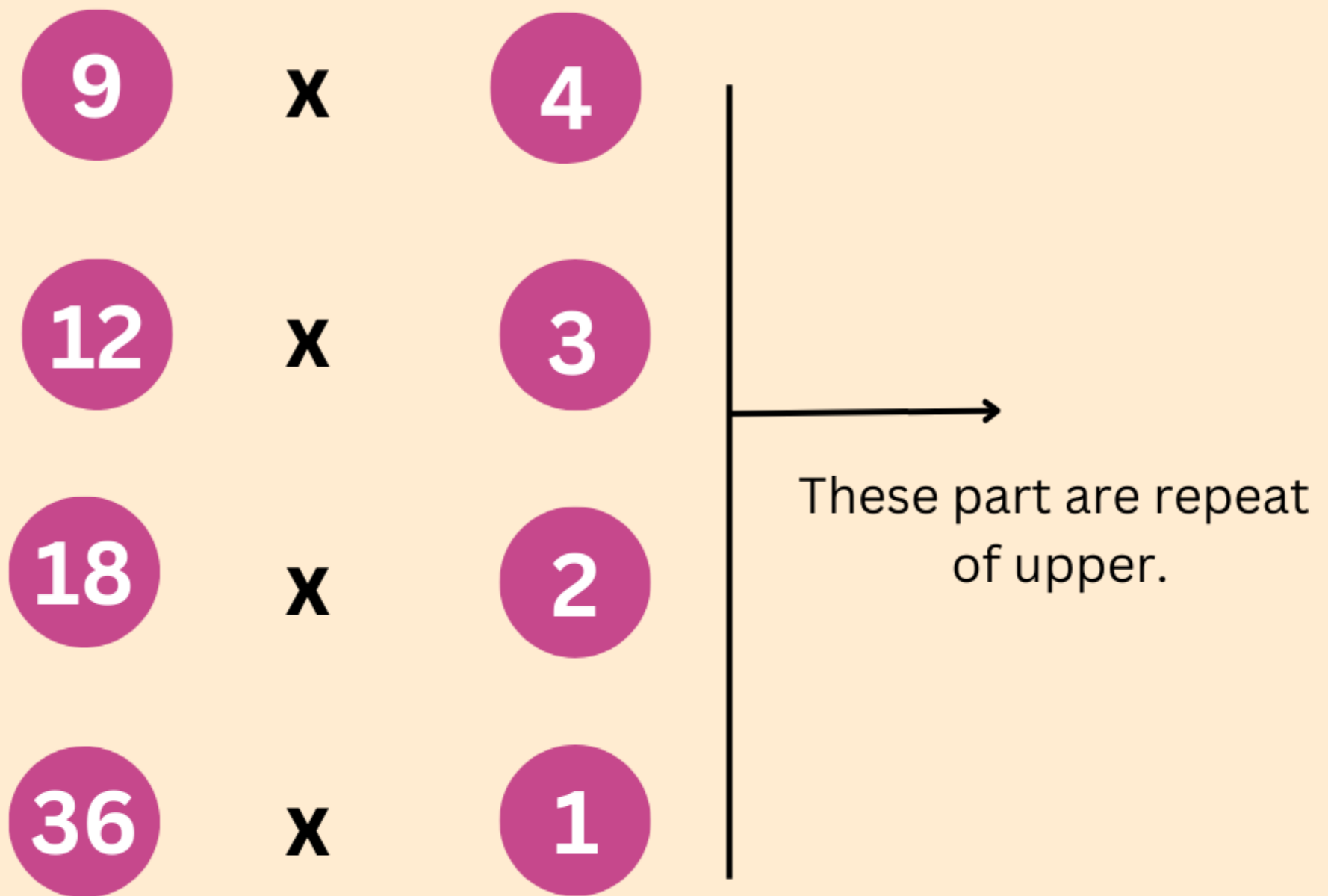
Space complexity: $O(1)$

But it can be done in  
 $\text{Sqrt}(N)$

**$N = 36$**

|   |   |    |
|---|---|----|
| 1 | x | 36 |
| 2 | x | 18 |
| 3 | x | 12 |
| 4 | x | 9  |
| 6 | x | 6  |

These part are repeat  
of lower.



- **$\text{sqrt}(N)$  = repeat part**

e.g:  $\text{sqrt}(36) = 6$

```

9  #include <bits/stdc++.h>
10 using namespace std;
11
12 bool isPrimeNo(int N)
13 {
14     int cnt = 0;
15     for (int i = 1; i * i <= N; i++)
16     {
17         if (N % i == 0)
18         {
19             cnt++;
20             if(N/i != i)
21             {
22                 cnt++;
23             }
24         }
25     }
26     if(cnt == 2)
27     {
28         return true;
29     }
30     return false;
31 }
32
33 int main()
34 {
35     int N = 11;
36     if (isPrimeNo(N))
37     {
38         cout << "It's a prime number." << endl;
39     }
40     else
41     {
42         cout << "It's not a prime number." << endl;
43     }
44
45     return 0;
46 }
47

```

It's a prime number.

**Time complexity:  $\sqrt{N}$**

**Space complexity:  $O(1)$**



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